

Yerman J. Merel G.

Panamá City, Panamá | (+507) 6295 – 9441 | ymerel_02@outlook.com | [Portfolio](#) | [LinkedIn](#)

Mechanical Engineer with a focus on Systems Engineering (MBSE). Specialist in transforming technical operations into profitable workflows through computational modeling and cost analysis. Experienced in international underground infrastructure projects and designing operational architectures for R&D startups.

EDUCATION

Technology University of Panama

Bachelor of Science in Mechanical Engineering

Relevant Coursework: Numerical Computations, Feedback Control Systems

Panama City, Panamá

Jan 2018 – Ago 2025

PROFESSIONAL EXPERIENCE

MONCAN

Co-Founder & Director of Systems Engineering

Panama City, Panamá

Jan 2025 – Jul 2025

- Implemented MBSE methodologies to define the company's operational architecture, ensuring full alignment between technical R&D requirements and the core business model.
- Designed and integrated digital workflows within the customer acquisition system, significantly reducing operational friction during early-stage lead generation while maintaining end-to-end prospect traceability.
- Developed cost-analysis models and systemic frameworks to optimize resource allocation, transforming complex technical operations into scalable and profitable business workflows

Herrenknecht Tunnelling Services Panama Corp.

Project Engineer

Panama City, Panamá

Jan 2025 – Jul 2025

- Designed and documented the global SOP for race replacement in the Main Drive of EPB TBM machines.
- Performed structural calculation validation and functional analysis of mechanical systems, ensuring strength and safety in high-demand industrial environments according to site conditions.
- Guaranteed 100% regulatory compliance and documentary traceability, mitigating contractual fine risks for the Lima Metro and Paladise Tunnel projects.

Projects: Hudson River Project (USA), Railway Tunnel Perú

China Railway Tunnel Group Co., LTD

Field Engineer of Tunnel Boring Machine and Superficial Facilities (*Intern*)

Panama City, Panamá

Aug 2024 – Jan 2025

- **Eliminated 100% of severe pipeline** incidents by designing and implementing corrective solutions including support systems for unstable terrain, pipe handling methods, and provisional sections for operational continuity.
- Developed standardized technical documentation adopted as an official reference by QC and Engineering teams, **reducing field errors by 20%** and improving cross-team efficiency.
- **Resolved critical failures** in high-pressure fluid systems (slurry/mortar) by designing isometric drawings and P&IDs, complying with **ASME B31** international standards.

TRAINING & COURSES

- **Design of Thermofluidic System** - Technology University of Panama (Aug 2023)
- **Fluid Technology, Hydraulic and Lubrication** - HERRENKNECHT AG (Jun 2025)

ADDITIONAL

Technical Skills: Advance in CAD (SolidWorks, AutoCAD, Inventor), ANSYS, Microsoft Office, SAP Business Suite

Languages: Spanish (Native); English (C1 - Full professional proficiency); German (A2 - Elementary Proficiency)

Awards: Finalist – National Scientific Initiation Conference (2023) / Resonant Wind Turbine Mathematical Model.